

Automatic Rigidity for Collatz-Invariant 3-Automatic Sequences

Muteki

Independent Researcher (Japan)

Contact: via the author's Ameba blog

Lean 4 formally verified version (Cobham's theorem used as an external assumption)

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Abstract

Let $a = (a_n)_{n \geq 0}$ be a sequence over a finite alphabet. We prove that if a is 3-automatic and satisfies

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0),$$

then a is constant on the positive integers, while a_0 remains free. Equivalently, every finite-alphabet 3-automatic coloring invariant under the shortcut Collatz map is trivial on the positive integers. In the first stage of the proof, we show that the 3-kernel is closed under the two binary decimations $u_n \mapsto u_{2n}$ and $u_n \mapsto u_{2n+1}$, yielding $K_2(a) \subseteq K_3(a)$. Thus a is both 2-automatic and 3-automatic, and Cobham's theorem implies ultimate periodicity. In the second stage, two identities satisfied by the odd-indexed subsequence $b_n = a_{2n+1}$ eliminate every nontrivial eventual period, forcing b , and hence a on the positive integers, to be constant. As a consequence, the Collatz conjecture is equivalent to 3-automaticity of the characteristic sequence of the basin of 1. This equivalence is a reformulation of the Collatz conjecture, not a proof of it.

The main proof chain and the principal corollaries were formally verified in Lean 4 / Mathlib. Cobham's theorem itself was not reproved inside Lean; instead, it is supplied to the proof chain as an explicit external assumption. In the final axiom audit, `sorryAx` did not appear; the remaining dependencies were limited to `propext`, `Classical.choice`, and `Quot.sound`.

Keywords: Collatz conjecture; automatic sequence; Cobham's theorem; finite automata; k -kernel; invariant set; base 3; Lean 4.

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Author's note: how this study arose

This work began with the question of whether the automatic-rigidity result previously obtained in base 2 could be extended to base 3. I am not a professionally trained mathematician; I have been exploring the neighborhood of the Collatz conjecture through sustained discussion with a contemporary conversational AI. My role has been to raise questions, choose directions, doubt novelty claims and logical gaps, and repeatedly request verification.

In base 3, rather than replacing the primitive-root arithmetic lemma used in the base-2 case, we found that Collatz invariance itself makes the 3-kernel closed under binary decimation. This reverse-direction kernel transfer is the main new structural feature of the present argument.

Before fixing the theorem statement, we also surveyed literature on finite automata, recognizable sets, self-generating sets, Collatz transducers, Büchi arithmetic, and reachability,

including the possibility that an equivalent statement might already exist under different terminology. Within the literature search carried out so far, we did not locate a prior result equivalent to the theorem proved here. This is necessarily a provisional conclusion dependent on the scope of the search, and it does not prove that no equivalent general theorem exists under different language.

1 Introduction

Define the shortcut Collatz map $T : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ by

$$T(2n) = n, \quad T(2n+1) = 3n+2 \quad (n \geq 0). \quad (1)$$

On the positive integers this is the usual Collatz map with one division by 2 built into the odd step. For general background on the $3x+1$ problem, see Lagarias [10].

Let Δ be a finite alphabet. A coloring $a : \mathbb{N}_0 \rightarrow \Delta$ is T -invariant when

$$a_n = a_{T(n)} \quad (n \geq 0).$$

Using (1), this is equivalent to

$$a_{2n} = a_n, \quad a_{2n+1} = a_{3n+2} \quad (n \geq 0). \quad (2)$$

Finite-state descriptions of Collatz structure have been studied for many years. Shallit and Wilson [11] treated predecessor families with a fixed number of odd Collatz steps using finite automata, while Stérin [12] constructed regular expressions for ancestor sets with a bounded number of odd steps. Stérin and Woods [13] showed that the Collatz process embeds a base-3 to base-2 conversion algorithm. Caucal and Rispal [3] studied transducer representations of Collatz-type maps and their powers. Dhiman and Pandey [5, 6] proved non-definability results in Büchi arithmetic for two-variable transition and reachability relations associated with generalized Collatz maps.

The question considered here is a different, one-dimensional rigidity problem: can a finite coloring that is nonconstant on the positive integers and invariant along every shortcut Collatz edge be 3-automatic? The answer is no.

Theorem 1.1 (Automatic rigidity in base 3; ARC_3). *Let Δ be a finite alphabet and let $a = (a_n)_{n \geq 0} \in \Delta^{\mathbb{N}_0}$ be a 3-automatic sequence. Assume (2) for every $n \geq 0$. Then there exists $c \in \Delta$ such that*

$$a_n = c \quad (n \geq 1).$$

The value a_0 is arbitrary.

Theorem 1.1 makes no assumption about the truth of the Collatz conjecture. It is a finite-state rigidity theorem for the map (1). In the functional graph of T , equation (2) says that the coloring is constant along every directed edge; the theorem states that finite-state base-3 structure cannot preserve different colors on distinct positive invariant components.

The proof separates into two mechanisms. Section 3 shows that the 3-kernel is closed under binary decimation, from which $K_2(a) \subseteq K_3(a)$ follows, and then applies Cobham's theorem. Section 4 uses two residual Collatz identities on the odd-indexed subsequence to eliminate the eventual period supplied by Cobham's theorem.

2 Automatic sequences and kernels

For an integer $k \geq 2$, the k -kernel of a sequence $u = (u_n)_{n \geq 0}$ is

$$K_k(u) := \{(u_{k^e n + r})_{n \geq 0} : e \geq 0, 0 \leq r < k^e\}.$$

For base- k digit representations we use the standard most-significant-digit-first convention, with no leading zeros and with 0 represented by the one-digit word 0. A sequence over a finite alphabet is k -automatic if and only if its k -kernel is finite [1].

We also use the sequence form of Cobham's theorem [1, 4]: if $k, \ell \geq 2$ are multiplicatively independent and a finite-alphabet sequence is both k -automatic and ℓ -automatic, then it is ultimately periodic.

In the base-2 ARC result, the fact that 2 is a primitive root modulo 3^j was used to embed the 3-kernel of the odd-indexed subsequence into the finite 2-kernel. The situation in base 3 is different. We now show that the entire 3-kernel is directly closed under binary decimation.

3 From the 3-kernel to the 2-kernel

For a sequence $u = (u_n)_{n \geq 0}$ define two decimation operators by

$$D_0 u = (u_{2n})_{n \geq 0}, \quad D_1 u = (u_{2n+1})_{n \geq 0}.$$

Lemma 3.1 (Binary-decimation closure of the 3-kernel). *Assume that a satisfies (2). Then for every $u \in K_3(a)$ and every $\varepsilon \in \{0, 1\}$,*

$$D_\varepsilon u \in K_3(a).$$

Proof. Let $u \in K_3(a)$. Then there exist $j \geq 0$ and $0 \leq r < 3^j$ such that

$$u_n = a_{3^j n + r}.$$

Fix $\varepsilon \in \{0, 1\}$ and put

$$d := \varepsilon 3^j + r.$$

Then $0 \leq d < 2 \cdot 3^j$ and

$$(D_\varepsilon u)_n = a_{3^j(2n+\varepsilon)+r} = a_{2 \cdot 3^j n + d}.$$

If d is even, the first identity in (2) gives

$$a_{2 \cdot 3^j n + d} = a_{3^j n + d/2}.$$

Since $0 \leq d/2 < 3^j$, this is an element of $K_3(a)$.

If d is odd, then

$$2 \cdot 3^j n + d = 2 \left(3^j n + \frac{d-1}{2} \right) + 1,$$

so the second identity in (2) gives

$$\begin{aligned} a_{2 \cdot 3^j n + d} &= a_{3^j n + (d-1)/2 + 2} \\ &= a_{3^{j+1} n + (3d+1)/2}. \end{aligned}$$

Moreover, $d \leq 2 \cdot 3^j - 1$, hence

$$0 \leq \frac{3d+1}{2} \leq 3^{j+1} - 1.$$

Thus $D_\varepsilon u \in K_3(a)$ in this case as well. $\square$

Proposition 3.2 (Kernel transfer). *If a is 3-automatic and satisfies (2), then*

$$K_2(a) \subseteq K_3(a).$$

In particular, a is also 2-automatic.

Proof. The sequence $a = (a_n)_{n \geq 0}$ itself belongs to $K_3(a)$. For arbitrary $e \geq 0$ and $0 \leq s < 2^e$, the subsequence

$$(a_{2^e n + s})_{n \geq 0}$$

is obtained from a by finitely many applications of D_0 and D_1 , following the binary digits of s . By Lemma 3.1, every intermediate sequence remains inside $K_3(a)$. Hence $K_2(a) \subseteq K_3(a)$. Since a is 3-automatic, $K_3(a)$ is finite, so $K_2(a)$ is finite as well. By the kernel characterization, a is 2-automatic. $\square$

Corollary 3.3. *Under the hypotheses of Theorem 1.1, the sequence a is ultimately periodic.*

Proof. By Proposition 3.2, a is both 2-automatic and 3-automatic. Since 2 and 3 are multiplicatively independent, Cobham's theorem implies that a is ultimately periodic. $\square$

This transfer runs in the opposite direction from the base-2 case. There, an arithmetic input was needed to embed subsequences indexed by powers of 3 into the 2-kernel. Here, because the coefficient 3 appearing in the odd Collatz branch naturally raises the level of the 3-kernel by one, closure under binary decimation is obtained directly.

4 Eliminating the eventual period

Define the odd-indexed subsequence by

$$b_n := a_{2n+1} \quad (n \geq 0). \quad (3)$$

By Corollary 3.3, b is also ultimately periodic.

Lemma 4.1. *For every $n \geq 0$,*

$$b_{4n} = b_{3n}, \quad b_{4n+2} = b_n.$$

Proof. Using (2) and (3),

$$b_{4n} = a_{8n+1} = a_{12n+2} = a_{6n+1} = b_{3n},$$

and

$$b_{4n+2} = a_{8n+5} = a_{12n+8} = a_{6n+4} = a_{3n+2} = b_n.$$

$\square$

Proposition 4.2. *If b is ultimately periodic and satisfies Lemma 4.1, then b is ultimately constant.*

Proof. Choose a threshold N_0 and a least positive eventual period P such that

$$b_{n+P} = b_n \quad (n \geq N_0).$$

For $x \in \mathbb{Z}/P\mathbb{Z}$, define $f(x)$ to be the common value of b_n for all sufficiently large $n \equiv x \pmod{P}$. Lemma 4.1 gives

$$f(4x) = f(3x), \quad f(4x+2) = f(x) \quad (4)$$

for all $x \in \mathbb{Z}/P\mathbb{Z}$.

First we show that P is odd. Put $d = P/\gcd(P, 4)$. Since $4d \equiv 0 \pmod{P}$, the second identity in (4) gives

$$f(x+d) = f(4x+4d+2) = f(4x+2) = f(x).$$

If $\gcd(P, 4) > 1$, then $0 < d < P$, so d would also be an eventual period, contradicting minimality of P . Hence $\gcd(P, 4) = 1$, so P is odd.

Next we show that $3 \nmid P$. Suppose instead that $3 \mid P$, and put $d = P/3$. For every x ,

$$f(4x+4d) = f(3x+3d) = f(3x) = f(4x).$$

Since 4 is invertible modulo P and $4d \equiv d \pmod{P}$, it follows that $f(y+d) = f(y)$ for every residue y , contradicting minimality of P . Therefore

$$\gcd(P, 6) = 1.$$

Because 3 and 4 are invertible modulo P , define affine permutations of $\mathbb{Z}/P\mathbb{Z}$ by

$$S(x) = \frac{3}{4}x, \quad F(x) = 4x + 2.$$

Equation (4) says that $f(Sx) = f(x)$ and $f(Fx) = f(x)$. Since S and F are permutations, f is also invariant under their inverses and all compositions. A direct calculation gives

$$SFS^{-1}F^{-1}(x) = x - \frac{1}{2}.$$

Thus f is invariant under translation by $-1/2$. Since P is odd, any integer u satisfying $2u \equiv -1 \pmod{P}$ has $\gcd(u, P) = 1$, so this translation acts transitively on $\mathbb{Z}/P\mathbb{Z}$. Hence f is constant. Therefore the least eventual period is $P = 1$, and b is ultimately constant. $\square$

Lemma 4.3. *If b is ultimately constant and satisfies $b_{4n+2} = b_n$ for every $n \geq 0$, then b is constant on all of $\mathbb{N}_0$.*

Proof. Put $F(n) = 4n + 2$. Iteration gives

$$F^k(n) = 4^k n + \frac{2(4^k - 1)}{3} \rightarrow \infty.$$

Since $b_{F(n)} = b_n$, we have $b_{F^k(n)} = b_n$. For sufficiently large k , $F^k(n)$ lies in the constant tail, so b_n equals the tail value. This holds for every $n \geq 0$. $\square$

Proof of Theorem 1.1. By Corollary 3.3, a is ultimately periodic, and hence the sequence b defined in (3) is ultimately periodic as well. Proposition 4.2 and Lemma 4.3 show that b is constant on all of $\mathbb{N}_0$. Let its common value be c .

Every integer $N \geq 1$ has a unique representation

$$N = 2^s(2m + 1)$$

with $s, m \geq 0$. Repeated use of the first identity in (2) gives

$$a_N = a_{2m+1} = b_m = c.$$

Thus a is constant on the positive integers.

Finally, $T(0) = 0$, and neither identity in (2) links a_0 to a positive index. For arbitrary $c, d \in \Delta$, the sequence defined by

$$a_0 = d, \quad a_n = c \quad (n \geq 1)$$

is 3-automatic and satisfies (2); hence a_0 is genuinely unconstrained. This sharpness example was also formally verified in Lean 4 (Section 8). $\square$

5 Consequences for Collatz-invariant 3-automatic sets

A subset $A \subseteq \mathbb{N}_0$ is called 3-automatic if its characteristic sequence is 3-automatic.

Corollary 5.1 (Invariant 3-automatic sets are trivial on $\mathbb{N}_{>0}$). *Let $A \subseteq \mathbb{N}_0$ be 3-automatic and suppose that*

$$n \in A \iff T(n) \in A$$

for every $n \geq 0$. Then either

$$A \cap \mathbb{N}_{>0} = \emptyset \quad \text{or} \quad A \cap \mathbb{N}_{>0} = \mathbb{N}_{>0}.$$

Membership of 0 is arbitrary.

Proof. Apply Theorem 1.1 to the characteristic sequence of A . $\square$

Corollary 5.2 (Nontrivial invariant sets are not 3-automatic). *Let $A \subseteq \mathbb{N}_{>0}$ satisfy*

$$n \in A \iff T(n) \in A \quad (n \geq 1).$$

If A is neither empty nor all of $\mathbb{N}_{>0}$, then its characteristic sequence is not 3-automatic.

Define the basin of 1 by

$$B := \{n \geq 1 : \exists k \geq 0, T^k(n) = 1\}. \tag{5}$$

We index its characteristic sequence on $\mathbb{N}_0$ and set $0 \notin B$.

Lemma 5.3. *The set B satisfies*

$$n \in B \iff T(n) \in B \quad (n \geq 0).$$

Proof. For $n = 0$, $T(0) = 0$ and both sides are false. Let $n \geq 1$. If $T(n) \in B$, then $T^k(T(n)) = 1$ for some $k \geq 0$, so $T^{k+1}(n) = 1$ and hence $n \in B$. Conversely, if $n \in B$ and $n \neq 1$, deleting the first step of a trajectory from n to 1 shows that $T(n) \in B$. When $n = 1$, we have $T(1) = 2$ and $T(2) = 1$, so $T(1) \in B$ as well. $\square$

Corollary 5.4 (A 3-automaticity criterion for the Collatz conjecture). *The Collatz conjecture on the positive integers holds if and only if the characteristic sequence of B is 3-automatic.*

Proof. If the Collatz conjecture holds, then $B = \mathbb{N}_{>0}$, whose characteristic sequence is automatic in every base. Conversely, if the characteristic sequence of B is 3-automatic, then Lemma 5.3 and Corollary 5.1 imply that B is either empty or all of $\mathbb{N}_{>0}$. Since $1 \in B$, we obtain $B = \mathbb{N}_{>0}$, which is exactly the Collatz conjecture. $\square$

Remark 5.5. *This corollary is a reformulation of the Collatz conjecture, not a proof of it. Theorem 1.1 does not show that the characteristic sequence of B is 3-automatic. Rather, it says that if the Collatz conjecture is false, then the full basin of 1 must possess genuinely non-3-automatic global complexity.*

6 Relation to previous work and novelty audit

The results of Shallit and Wilson [11] and Stérin [12] concern regularity / automaticity of predecessor or ancestor sets with a bounded number of odd steps. They provide finite-state descriptions of particular inverse-orbit families, not a classification of arbitrary T -invariant 3-automatic colorings.

Garth and Gouge [7], and Kärki, Lacroix, and Rigo [8], studied recognizability / automaticity of minimal closures generated from finitely many affine maps. In particular, the latter showed under technical hypotheses that self-generating sets involving multiplicatively independent coefficients are not recognizable in any base. Their objects, however, are minimal forward closures generated from finite seeds, whereas the present condition

$$n \in A \iff T(n) \in A$$

expresses two-sided orbit saturation of a pre-existing set. The preservation problems for recognizable sets under affine / linear transformations studied by Klarner [9] and Boigelot [2] likewise provide important background, but do not directly imply Theorem 1.1.

Stérin and Woods [13] made explicit the interaction between bases 2 and 3 in the Collatz process, while Caucal and Rispal [3] constructed transducers computing Collatz-type maps and their powers. The present work does not compute the map itself; it classifies finite-valued 3-automatic invariants that remain constant along every edge of the map.

Dhiman and Pandey [5, 6] proved that two-variable transition and reachability relations for generalized Collatz maps are not definable in suitable Büchi arithmetic / finite-automata frameworks, using Cobham–Seménov / Cobham-type base rigidity. Our object is instead a one-variable finite coloring $a(n)$ satisfying $a(T(n)) = a(n)$. Neither statement directly implies the other.

For the novelty audit, Theorem 1.1 was searched not only in sequence form, but also as the recognizable-invariant-set formulation of Corollary 5.1, as a component-coloring statement for the Collatz functional graph, and as a base-3 finite-state unary-predicate statement. Within the literature examined as of September 2026, we did not find a prior theorem explicitly asserting that every finite-alphabet 3-automatic coloring invariant under the shortcut Collatz map is constant on the positive integers. This statement does not claim exhaustive nonexistence of prior art; an equivalent theorem under different terminology or as an immediate consequence of a more general result cannot be ruled out completely.

7 Comparison with the base-2 result and prospects for generalization

The base-2 and base-3 proofs reach the same conclusion through different kernel-transfer mechanisms.

	Base 2	Base 3
Starting point	a is 2-automatic	a is 3-automatic
Main transfer	Embed the 3-kernel of the odd subsequence into $K_2(a)$	Prove $K_2(a) \subseteq K_3(a)$ directly
Arithmetic input	2 is a primitive root modulo 3^j	None
Sequence before Cobham	Odd subsequence b	The sequence a itself
Period collapse	$b_{4n} = b_{3n}, b_{4n+2} = b_n$	Reuse the same identities

The direct closure in base 3 occurs because the coefficient 3 in the odd branch

$$2n + 1 \mapsto 3n + 2$$

fits exactly with the hierarchy of the 3-kernel. This suggests that the same proof will not extend mechanically to an arbitrary base q .

A natural next problem is to characterize those bases $q \geq 2$ for which

$$q\text{-automatic} + T\text{-invariant} \implies \text{constant on } \mathbb{N}_{>0}$$

holds. Studying bases such as 5, 7, 6, 10 individually may separate two ingredients: (i) a kernel-transfer mechanism forcing automaticity in another base, and (ii) residual affine symmetries strong enough to collapse the eventual period after applying Cobham's theorem.

8 Formal verification in Lean 4

The main proof chain and the principal consequences of this paper were formally verified in Lean 4 / Mathlib. The formalization reuses the automaticity bridges, period-collapse machinery, and main-theorem components developed for the base-2 case, while adding the base-3-specific kernel transfer as a new component.

Files and verified scope

File	Main verified content
ARC3KernelTransfer.lean	Closure of the 3-kernel under even and odd binary decimation; $K_2(a) \subseteq K_3(a)$; transfer from finite 3-kernel to finite 2-kernel.
ARC3Main.lean	Connection from 3-automaticity to 2-automaticity and the ARC_3 main theorem by reusing the already verified base-2 ARC core. Includes an entry point from MSD automaticity.
ARC3Audit.lean	An explicit formal counterpart of the paper’s route “2-automatic and 3-automatic $\Rightarrow$ Cobham $\Rightarrow$ ultimately periodic”, plus bridges for standard MSD conventions, no leading zeros, and representing 0 by the one-digit word [0].
ARC3Consequences.lean	Sharpness example showing that a_0 is genuinely free; triviality of invariant 3-automatic sets; T -invariance of the basin of 1; equivalence between the Collatz conjecture and 3-automaticity of that basin.

The base-3-specific core is the transfer

$$K_2(a) \subseteq K_3(a).$$

Boundary cases including $j = 0$, residue $r = 0$, the maximal residue, parity splits involving natural-number division, and induction over binary residues were adversarially audited and compiled successfully.

Treatment of Cobham’s theorem

Cobham’s theorem itself is not reproved inside this formalization. As in the base-2 development, a standard form of Cobham’s theorem is represented by `ARCCobhamPrinciple` and passed to the main proof as an explicit external hypothesis. Thus “formally verified in Lean” means precisely

assuming the standard Cobham theorem externally, Lean 4 verifies the remaining proof chain.

Axiom audit

An audit equivalent to `#print axioms` was performed on the main theorem and principal corollaries. In the final state, `sorryAx` does not appear; the dependencies are limited to `propext`, `Classical.choice`, and `Quot.sound`. These are standard dependencies associated with ordinary classical mathematics and quotient constructions in Lean / Mathlib, and are distinct from `sorryAx`, which would signal an unproved hole.

Representative theorem names confirmed in the final audit include:

```
arc3_two_kernel_subset_three_kernel
arc3_main_standard_msd_form_of_cobham
arc3_zero_value_is_genuinely_free
arc3_invariant_automatic_set_trivial
arc3_collatz_iff_basin_three_automatic
```

In addition to per-file `lake env lean` checks, the development was verified by Lake builds including dependencies. The remaining diagnostics are limited to linter / style warn-

ings such as empty lines, unused `simp` arguments, and deprecated notation; they are not logical errors.

Disclosure of generative-AI assistance

Generative AI (OpenAI ChatGPT, GPT-5.6 Sol) was used in preparing this manuscript for literature-search assistance, theorem formulation, adversarial checking of proof steps, editorial drafting, assistance with Lean 4 proof-code development, diagnosis of compilation errors, and post-formalization auditing. The AI system is not an author and is not used as a bibliographic source. Responsibility for the mathematical claims, references, formalization, and final manuscript rests with the human author.

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